Practice workbook · Using the sine rule, including the two-answer case

Using the sine rule to find a side or an angle, including recognising when an angle can be acute or obtuse.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-09

1. Question 3 · part whole · 5 marks

2024 NOV · Paper 01

Worked-example retry · Attempt part whole only.

3 Triangle ABC is such that

$AC = 10\,\text{cm}$ $BC = 7\text{cm}$ angle $CAB = 25°$

Given that angle ABC is obtuse,

find, in cm to one decimal place, the length of AB

(5)

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1. Question 3 · part whole · 5 marks

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Worked-example retry · Attempt part whole only.

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Question 3 continued

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(Total for Question 3 is 5 marks)

1. Question 3 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 3 · part b.i · 3 marks

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Further practice · Attempt part b.i only.

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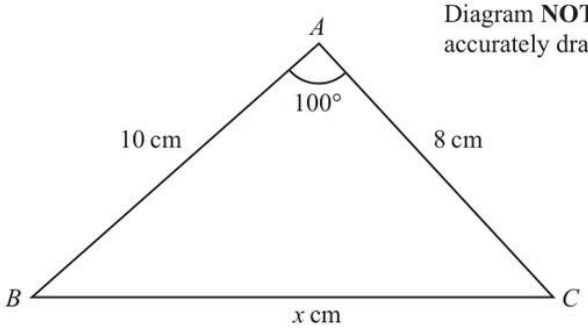


Figure 2

Figure 2 shows triangle ABC where

$$AB = 10 \text{ cm}, \quad AC = 8 \text{ cm}, \quad BC = x \text{ cm} \quad \text{and} \quad \angle BAC = 100^\circ$$

(a) Find, to 3 significant figures, the value of x

(2)

(b) Find, in degrees to one decimal place, the size of

(i) angle ABC (ii) angle ACB

(3)

The bisector of angle ABC meets AC at the point M

(c) Find the area, in cm^2 to 3 significant figures, of triangle BMC.

(4)

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P 7 4 0 9 8 A 0 6 3 2

2. Question 3 · part b.i · 3 marks · Working and answer

Working and answer

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3. Question 3 · part c.i · 0 marks

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Further practice · Attempt part c.i only.

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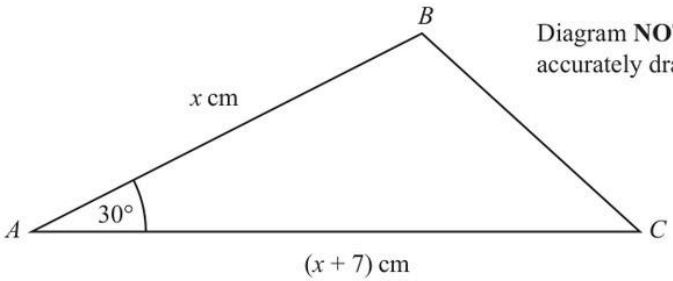


Figure 1

Figure 1 shows triangle ABC where

$AB = x$ cm $\quad AC = (x + 7)$ cm $\quad \angle BAC = 30°$

The area of triangle $ABC = 36$ cm^2

(a) Show that $x = 9$ **(3)**

(b) Find, in cm to 3 significant figures, the length of BC **(2)**

(c) Find, in degrees to one decimal place, the size of

(i) $\angle ABC$

(ii) $\angle ACB$ **(3)**

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3. Question 3 · part c.i · 0 marks · Working and answer

Working and answer

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4. Question 3 · part c.ii · 3 marks

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Further practice · Attempt part c.ii only.

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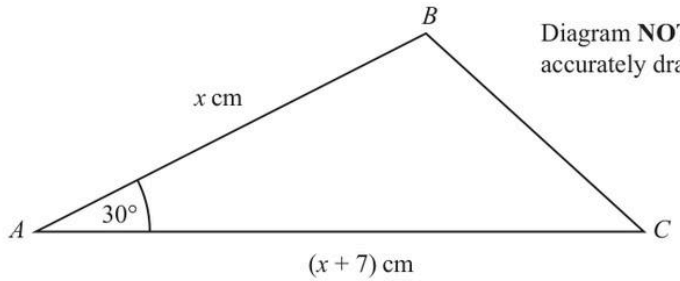


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P 7 4 0 9 9 A 0 6 3 2

4. Question 3 · part c.ii · 3 marks

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Further practice · Attempt part c.ii only.

Question 3 continued

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(Total for Question 3 is 8 marks)

4. Question 3 · part c.ii · 3 marks · Working and answer

Working and answer

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5. Question 2 · part whole · 3 marks

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Further practice · Attempt part whole only.

2 In ΔABC,

$$\angle BAC = 50° \qquad AB = 10\,\text{cm} \qquad BC = 9\,\text{cm}$$

Given that $\angle BCA = x°$

find the two possible values, to one decimal place, of x

(3)

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5. Question 2 · part whole · 3 marks

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Further practice · Attempt part whole only.



Question 2 continued

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(Total for Question 2 is 3 marks)

5. Question 2 · part whole · 3 marks · Working and answer

Working and answer

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6. Question 4 · part a · 4 marks

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Further practice · Attempt part a only.

4 Triangle ABC is such that

$AB = 4.3$ cm $\quad BC = 5.9$ cm $\quad$ angle $BCA = 29°$

(a) Find in degrees, to one decimal place, the two possible values for angle CAB

(4)

(b) Hence, find in cm, to 3 significant figures, the shortest possible length of AC

(3)

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P 7 6 5 0 7 A 0 8 3 2

6. Question 4 · part a · 4 marks · Working and answer

Working and answer

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7. Question 4 · part b · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

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(a) Find in degrees, to one decimal place, the two possible values for angle CAB

(4)

(b) Hence, find in cm, to 3 significant figures, the shortest possible length of AC

(3)

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7. Question 4 · part b · 3 marks

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Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 7 marks)

7. Question 4 · part b · 3 marks · Working and answer

Working and answer

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8. Question 3 · part whole · 7 marks

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Further practice · Attempt part whole only.

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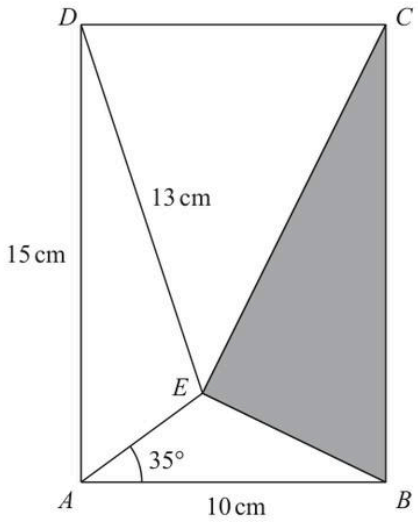


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a rectangle $ABCD$ with $AB = 10$ cm and $AD = 15$ cm.
E is the point inside the rectangle such that $DE = 13$ cm and angle $BAE = 35°$

Given that angle AED is obtuse,

find the area, in cm^2 to one decimal place, of triangle BCE.

(7)

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8. Question 3 · part whole · 7 marks

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Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 7 marks)

8. Question 3 · part whole · 7 marks · Working and answer

Working and answer

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9. Question 9 · part c · 4 marks

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Further practice · Attempt part c only.

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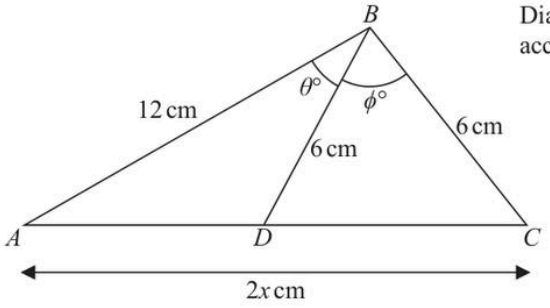


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle ABC with $AB = 12\,\text{cm}$, $BC = 6\,\text{cm}$ and $AC = 2x\,\text{cm}$.

The point D is the midpoint of AC and $BD = 6\,\text{cm}$.

$\angle ABD = \theta°$ and $\angle DBC = \phi°$ where $\theta \neq 0$ and $\phi \neq 0$

(a) Show that $\cos ADB = \dfrac{x^2 - 108}{12x}$ **(2)**

(b) Hence, or otherwise, show that $AC = 6\sqrt{6}\,\text{cm}$. **(4)**

(c) Show that $\sin(\theta° + \phi°) = \sin\phi°$ **(4)**

(d) Hence show that $\theta = 180 - 2\phi$ **(2)**

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9. Question 9 · part c · 4 marks · Working and answer

Working and answer

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10. Question 9 · part d · 2 marks

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Further practice · Attempt part d only.

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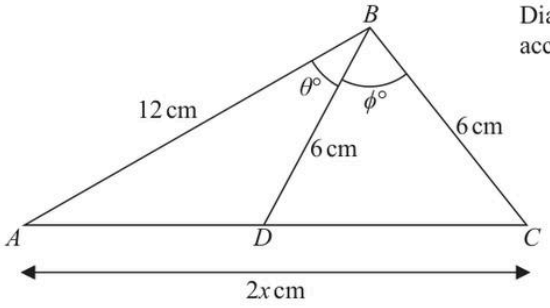


Diagram **NOT** accurately drawn

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(c) Show that $\sin(\theta° + \phi°) = \sin\phi°$ **(4)**

(d) Hence show that $\theta = 180 - 2\phi$ **(2)**

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10. Question 9 · part d · 2 marks

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Question 9 continued

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10. Question 9 · part d · 2 marks

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Further practice · Attempt part d only.

Question 9 continued

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10. Question 9 · part d · 2 marks

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Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 12 marks)

10. Question 9 · part d · 2 marks · Working and answer

Working and answer

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11. Question 6 · part a.i · 0 marks

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Further practice · Attempt part a.i only.

6 In triangle ABC, $AC = x$ cm, $AB = (x + 3)$ cm and $\angle ABC = 30°$

Given that $\angle ACB = \theta°$ where $0 < \theta < 90$

(a) show that

(i) $\sin\theta° = \dfrac{x + 3}{2x}$

(ii) $\cos\theta° = \dfrac{\sqrt{3x^2 - 6x - 9}}{2x}$

(5)

Given that the size of $\angle BAC$: the size of $\angle ABC = 7 : 2$

(b) find the exact value of x

Give your answer in the form $a + a\sqrt{b}$ where a and b are prime numbers.

(5)

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11. Question 6 · part a.i · 0 marks · Working and answer

Working and answer

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12. Question 6 · part b · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

6 In triangle ABC, $AC = x$ cm, $AB = (x + 3)$ cm and $\angle ABC = 30°$

Given that $\angle ACB = \theta°$ where $0 < \theta < 90$

(a) show that

(i) $\sin\theta° = \dfrac{x + 3}{2x}$

(ii) $\cos\theta° = \dfrac{\sqrt{3x^2 - 6x - 9}}{2x}$

(5)

Given that the size of $\angle BAC$: the size of $\angle ABC = 7 : 2$

(b) find the exact value of x

Give your answer in the form $a + a\sqrt{b}$ where a and b are prime numbers.

(5)

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12. Question 6 · part b · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 6 continued

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12. Question 6 · part b · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 6 continued

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12. Question 6 · part b · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 10 marks)

12. Question 6 · part b · 5 marks · Working and answer

Working and answer

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13. Question 6 · part b · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 Given that $\dfrac{a+\sqrt{5}}{\sqrt{5}-2} = 11 + 5\sqrt{5}$

(a) without using a calculator, find the value of a
Show your working clearly.

(2)

Triangle PQR is such that

$PR = (x+3)\,\text{cm}$ $QR = x\,\text{cm}$ angle $QPR = 30°$ angle $PQR = 45°$

(b) Show that $x = 3 + 3\sqrt{2}$

(3)

Given that $\sin 105° = \dfrac{\sqrt{6}+\sqrt{2}}{4}$ and that the area of triangle PQR is $A\,\text{cm}^2$

(c) find the exact value of A in the form $\dfrac{9}{8}\left(p\sqrt{6} + q\sqrt{2} + r\sqrt{3} + s\right)$

where p, q, r and s are integers.

(3)

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13. Question 6 · part b · 3 marks · Working and answer

Working and answer

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14. Question 9 · part b · 5 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

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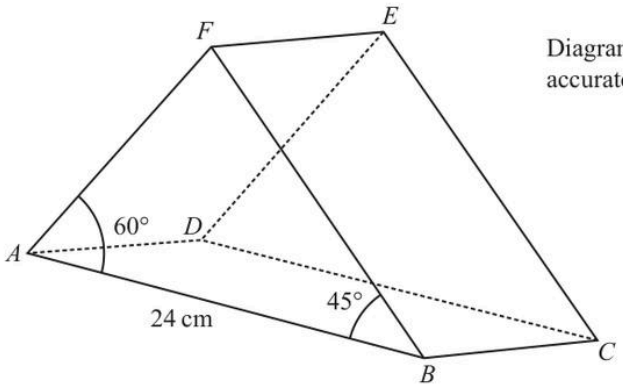


Figure 5

Figure 5 shows a right triangular prism $ABCDEF$ where $ABCD$ is a rectangle.

$AF = DE$ $BF = CE$ $AD = FE = BC$ $AB = DC = 24$ cm

$\angle ABF = \angle DCE = 45°$ $\angle BAF = \angle CDE = 60°$

Using a formula from page 2,

(a) show that $\sin AFB = \dfrac{\sqrt{2} + \sqrt{6}}{4}$

(3)

Without using a calculator,

(b) show that $BF = 12\left(3\sqrt{2} - \sqrt{6}\right)$ cm

(5)

The angle between the plane AEB and the plane $ABCD$ is 65°

(c) Find, in cm to 2 significant figures, the length of EF

(3)

(d) Find, in degrees to one decimal place, the size of the angle between the line CF and the plane $ABCD$

(4)

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14. Question 9 · part b · 5 marks · Working and answer

Working and answer

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